

CHRONO-Fair: Anytime-Valid Counterfactual Flip Monitoring for Clinical Machine Learning

ANONYMOUS AUTHOR(S)

Clinical prediction models are deployed under regulatory regimes that require fairness monitoring after release. The FDA 2024 Predetermined Change Control Plan guidance and EU AI Act Articles 10 and 17 both expect continuous post-market bias surveillance. The auditing tools in current use compute a group fairness metric on a fixed snapshot of predictions. They then report a point estimate against a threshold, such as the 0.8 rule. A point estimate, however, hides whether the verdict it produces is stable. On the real Texas-100X discharge dataset (925 128 records), the worst-group true-positive-rate verdict for race reads “fair” at a point value of 0.824. The same verdict flips to “unfair” in 4 of 20 hospital-network resamples, a Verdict Flip Rate of 0.20. No tool in clinical use reports when such a verdict flips during a live patient stream, nor whether the flip is driven by biased data or by insufficient data.

This gap has three parts that prior work treats in isolation. A bootstrap confidence interval reveals that a verdict is uncertain, but it is computed once and is silent during deployment. The Verdict Flip Rate quantifies how often the verdict flips across resamples, but it is a scalar with no time axis and no statistical false-alarm control. Drift detectors such as ADWIN operate sequentially, but they monitor a single metric and provide no anytime-valid guarantee under repeated inspection. None of the three attributes a detected flip to a cause that a clinical governance team can act on.

This paper presents CHRONO-Fair, a simulation-based monitoring prototype that closes the three parts of the gap with four coordinated estimators. A Flip Hazard survival object recasts the scalar flip rate as a group-conditional Kaplan-Meier curve with log-rank tests. An anytime-valid e-process raises a statistically controlled alarm at any inspection time, with step-wise Benjamini-Hochberg control across intersectional cells. An ensemble entropy decomposition splits a detected flip into an aleatoric (data) cause and an epistemic (sample-size) cause. A Wasserstein-1 Regression Counterfactual Allocation Parity statistic extends the analysis to length-of-stay regression. On the real Texas-100X verdict data, the confidence interval and the Verdict Flip Rate agree on all 12 audited metric-attribute combinations, and both isolate the single unstable verdict that the point estimate misses. On synthetic streams calibrated to the same dataset, the e-process detects an injected drift in 50 of 50 runs at a median delay of 848 patients, against 21 of 50 for ADWIN, while holding the empirical false-alarm rate at or below the nominal level. All findings are simulation-based or based on stored real-data summary statistics, and require external validation on live electronic health record streams.

Additional Key Words and Phrases: algorithmic fairness, clinical machine learning, anytime-valid inference, fairness drift, post-deployment monitoring, length-of-stay prediction

1 INTRODUCTION

Healthcare prediction models can pass a one-off fairness audit at the time of deployment and develop subgroup gaps later. Obermeyer et al. [82] documented racial bias in a widely used care-management algorithm. Seyyed-Kalantari et al. [100] reported underdiagnosis of female and minority patients by chest radiograph classifiers. Habib et al. [43] and Wong et al. [116] showed that the Epic Sepsis Model underperformed in external validation. Davis et al. [28] analysed eleven years of VA cohort data and observed that fairness gaps drift after deployment, frequently in the direction of widening.

Regulatory bodies responded. The U.S. FDA issued its Predetermined Change Control Plan final guidance in December 2024 [105]. Article 17 of the EU AI Act, in force from August 2027 for high-risk medical AI, requires continuous post-market bias monitoring [36]. The DECIDE-AI [107], TRIPOD+AI [25] and CONSORT-AI [68] reporting standards now

expect lifecycle evaluation. McCradden et al. [75] and Eaneff et al. [35] have argued for “algorithmic stewardship” as a regulatory category.

The methodological toolchain has not caught up. Open-source audit packages (Aequitas, Fairlearn, AIF360, Microsoft Responsible AI Dashboard) operate in batch mode on snapshot predictions. The online fair-learning line of work [49, 50, 81] demonstrates streaming feasibility on COMPAS and credit data, not on EHR streams. Anytime-valid inference [42, 47, 92, 115] has been applied to A/B-test monitoring and calibration drift, but not to fairness drift at intersectional granularity. Predictive-multiplicity work [12, 26, 73] characterises pipeline instability but is framed as reliability rather than fairness.

Prior work by some of the present authors introduced the Verdict Flip Rate (VFR), a scalar that quantifies how often a fairness verdict flips between resamples of a hospital-network bootstrap [6]. VFR captures something the canonical group metrics miss, namely the instability of a fairness verdict under stochastic pipeline choices. The metric is, however, scalar, batch, single-threshold, and gives no actionable indication of when, where, or why a gap arose.

Contributions. This paper makes four contributions, each presented as a simulation-based prototype.

- (1) A Flip Hazard estimator that recasts the resampling-based VFR as a group-conditional survival object on the patient arrival index. Kaplan-Meier survival, Nelson-Aalen cumulative hazard, two-sample log-rank statistics, and a Restricted Mean Flip Time (RMFT) are reported per protected group.
- (2) An anytime-valid e-process monitor for the flip indicator with a shrunk GROW betting rule, a warm-up of 100 patients, and step-wise Benjamini-Hochberg multiple-testing control across intersectional cells. The empirical Type-I rate stays at or below the nominal α on synthetic streams.
- (3) An ensemble entropy decomposition that splits the per-cell flip mass into aleatoric and epistemic components. The split admits a clear governance interpretation: an aleatoric-dominated cell signals label or feature-pipeline bias, an epistemic-dominated cell signals sample-size scarcity.
- (4) A Regression Counterfactual Allocation Parity (RCAP) metric for length-of-stay regression. RCAP is the Wasserstein-1 distance between counterfactual rank distributions and is sensitive to allocation shifts that the group-MAE comparator does not register.

To our knowledge, no single published system combines these four estimators with intersectional anytime-valid control and an automated inspection report. The framework, the synthesiser, and the dashboard prototype are released as open-source code with the supplementary material.

2 RELATED WORK

2.1 Group, individual, and counterfactual fairness

Dwork et al. [34] defined individual fairness as a Lipschitz constraint. Two individuals within distance d on a task metric must receive outcome distributions within distance D . Hardt et al. [44] defined equalised odds as equal true-positive and false-positive rates across groups. Equal opportunity relaxes this to equal true-positive rate only. Chouldechova [23] proved that, when group base rates differ, a classifier cannot hold predictive parity, equal false-positive rate, and equal false-negative rate at once. Kleinberg et al. [62] proved a parallel result:

calibration and class-balanced error are jointly attainable only under equal base rates or perfect prediction. Pleiss et al. [87] showed the conflict persists for any non-trivial calibration constraint and quantified the achievable relaxation. Kusner et al. [64] defined counterfactual fairness as invariance of the prediction distribution under a do-intervention on the protected attribute in a structural causal model. Chiappa [22] restricted this to unfair causal paths only, leaving resolving paths intact. Nabi and Shpitser [78] constrained path-specific effects during estimation by a constrained likelihood. Kilbertus et al. [60] separated proxy discrimination from explanatory effects in a causal graph. Makhoul et al. [70] mapped 19 causal fairness notions to the data assumptions each one needs. The surveys of Verma and Rubin [108], Mehrabi et al. [76], Pessach and Shmueli [85], and Caton and Haas [18] catalogue 20 or more metrics. They report that the choice of metric, not the model, drives most reported disparity. On mitigation, Agarwal et al. [2] reduced fair classification to a sequence of cost-weighted subproblems solved by a saddle-point game. Donini et al. [31] added a fairness term to empirical risk minimisation with a generalisation bound. Calmon et al. [16] and Kamiran and Calders [56] reweighted or transformed training data before fitting. Madras et al. [69] and Lahoti et al. [65] learned representations that an adversary cannot use to recover the protected attribute. Wang et al. [113] bounded fairness violation when group labels are noisy. Kallus et al. [55] estimated disparity by combining a main dataset with an auxiliary dataset that carries the protected attribute.

2.2 Subgroup, multicalibration, and statistical-audit perspectives

Kearns et al. [58] showed that parity on coarse groups can hide large violations on structured subgroups, an effect they named fairness gerrymandering. Hébert-Johnson et al. [46] defined multicalibration as calibration that holds simultaneously on every subgroup in a computationally identifiable class. Kim et al. [61] produced multiaccuracy by post-processing, auditing for subgroups where the residual correlates with subgroup membership. Hashimoto et al. [45], Sagawa et al. [96], and Duchi and Namkoong [32] minimised worst-group loss through distributionally robust optimisation over an uncertainty set of group mixtures. Martínez et al. [72] characterised the accuracy-fairness frontier as a Pareto set and computed it through a multi-objective descent. Cherian and Candès [21] produced finite-sample confidence intervals for a fairness metric using bootstrap and rank-based statistics. Besse et al. [10] reported that the sampling variance of statistical parity can exceed the disparity itself when subgroup size is below a few hundred. Coston et al. [27] evaluated counterfactual risk under selective labels through inverse-propensity weighting. These works audit a fixed snapshot. None updates a fairness verdict sequentially as patients arrive.

2.3 Predictive multiplicity, model multiplicity, and stability

Breiman [15] observed that distinct models can attain near-identical loss while disagreeing on individuals. Marx et al. [73] formalised this as predictive multiplicity. They defined ambiguity as the fraction of inputs that receive a conflicting prediction from some model within an ϵ -loss Rashomon set. Watson-Daniels et al. [114] extended the measure to probabilistic outputs through the range of predicted scores. Semenova et al. [99] linked a large Rashomon set to the existence of a simpler accurate model. Dong and Rudin [30] computed variable-importance ranges across the Rashomon set rather than for one model. Black et al. [12] argued that model multiplicity gives a designer latitude to select a fairer model at no accuracy cost. Cooper et al. [26] showed that a measured group gap can be a variance artefact: averaging predictions across the Rashomon set shrank reported disparity. Miller and Blume [77] defined the empirical Decision Flip Rate as the fraction of patients whose binary decision changes

between bootstrap re-trainings. Black et al. [13] detected individual discrimination through an optimal-transport map between group-conditional input distributions. The Verdict Flip Rate of [6] measured how often the fairness verdict, not the prediction, flips between hospital-network resamples.

2.4 Uncertainty quantification and its decomposition

Depeweg et al. [29] decomposed predictive uncertainty into two terms. Aleatoric uncertainty is the mean of per-model entropies. Epistemic uncertainty is the entropy of the mean prediction minus that mean, equal to the mutual information between label and parameters. Hüllermeier and Waegeman [48] formalised the distinction and noted that only the epistemic term shrinks with more data. Lakshminarayanan et al. [66] estimated epistemic uncertainty from the disagreement of M independently trained networks. Gal and Ghahramani [39] approximated the same quantity through Monte Carlo dropout. Kendall and Gal [59] separated learned input-dependent variance from model uncertainty in a single loss. Ovadia et al. [83] reported that deep ensembles kept calibration error lowest among uncertainty methods under distribution shift. Alghamdi et al. [3] applied the aleatoric-epistemic split to discrimination, attributing the irreducible part to the data and the reducible part to the model class. Romano et al. [95] and Gibbs and Candès [41] produced prediction sets with a finite-sample coverage guarantee, the latter adapting the set width online under shift. Barber et al. [7] extended conformal coverage to non-exchangeable data with a bound that degrades in the total-variation distance from exchangeability. Angelopoulos and Bates [5] surveyed the coverage guarantees that the conformal family provides. Dutta et al. [33] and Calmon et al. [17] decomposed discrimination through partial information, separating the exempt component from the unfair component.

2.5 Anytime-valid inference, drift detection, online learning

Ville [109] proved that a non-negative supermartingale started at one crosses $1/\alpha$ with probability at most α . This single inequality underwrites every e-process. Wald [112] built the sequential probability ratio test, which stops as soon as a likelihood ratio leaves a fixed interval. Howard et al. [47] produced confidence sequences, intervals that cover the parameter simultaneously at every sample size with probability $1 - \alpha$. Waudby-Smith and Ramdas [115] estimated a bounded mean by betting. The wealth process $\prod_t (1 + \lambda_t(X_t - m))$ is a martingale under the null mean m . The GROW rule picks λ_t to maximise expected log-growth. Ramdas et al. [92] and Shafer [101] unified these results as game-theoretic statistics, where evidence equals the capital won against the null. Grünwald et al. [42] characterised the e-variable that is growth-optimal against a composite alternative. Johari et al. [53] applied always-valid p -values to continuous A/B-test monitoring, removing the penalty for repeated peeking. Podkopaev and Ramdas [88] tracked a deployed model's risk through a betting martingale on the loss sequence. Foster and Stine [38] introduced alpha-investing, which spends a false-discovery budget across an ordered hypothesis stream. Javanmard and Montanari [51] produced the LORD rules that control online false-discovery rate with a decaying memory. Bifet and Gavalda [11] introduced ADWIN, which keeps a variable-length window and drops the older half when two sub-window means differ by a Hoeffding bound. Gama et al. [40] taxonomised concept drift into sudden, gradual, and recurring types. Rabanser et al. [90] reported that dimensionality reduction followed by a two-sample test detected dataset shift most reliably. Iosifidis and Ntoutsi [49, 50] maintained a fairness constraint online by reweighting the loss as a stream evolves. None of these monitors a counterfactual flip indicator under intersectional multiple-testing control.

2.6 Healthcare AI fairness, clinical prediction, and deployment

Obermeyer et al. [82] traced the bias of a care-management algorithm to its label. The model predicted healthcare cost as a proxy for healthcare need. At a fixed risk score, Black patients carried more chronic conditions than White patients. Correcting the label raised the share of Black patients flagged for extra care from 17.7% to 46.5%. Vyas et al. [110] catalogued race-adjusted clinical formulas, including an eGFR correction that lowered estimated kidney disease severity for Black patients. Seyyed-Kalantari et al. [100] reported that chest radiograph classifiers had higher false-negative rates for female, Black, and publicly insured patients, with the largest underdiagnosis on intersectional subgroups. Wong et al. [116] externally validated the Epic Sepsis Model. They found an area under the receiver operating characteristic curve of 0.63, against the vendor-reported 0.76 to 0.83. Habib et al. [43] confirmed the external-validation gap and the high alert burden. Noseworthy et al. [80] tested an electrocardiogram deep model and found stable performance across race once age and sex were matched. Larrazabal et al. [67] showed that a sex-imbalanced training set lowered diagnostic accuracy for the under-represented sex. Coley et al. [24] reported that a suicide-risk model had lower sensitivity for Black and American Indian patients. Adam et al. [1] found that race-correlated language in clinical notes shifted downstream model recommendations. Pierson et al. [86] trained a pain model on patient-reported scores rather than radiologist labels and explained a larger share of the racial pain gap. Davis et al. [28] tracked an in-use model across eleven years of VA data and observed subgroup performance drift after deployment. Rajkomar et al. [91] and Chen et al. [19] set out fairness desiderata for clinical machine learning. Singh et al. [102] argued that a deployed clinical model needs continual rather than one-off evaluation. Finlayson et al. [37] catalogued the dataset-shift modes that degrade clinical models, including case-mix and practice-pattern change. Vasey et al. [107] and Collins et al. [25] issued the DECIDE-AI and TRIPOD+AI reporting standards, both of which now require subgroup performance reporting. Sculley et al. [98] and Breck et al. [14] documented the monitoring debt that accrues when a model ships without a test harness. The MIMIC-IV [54] and eICU [89] databases supply the public critical-care streams used for external validation. Texas-100X [52] provides 925 128 hospital discharge records with the demographic mix this paper’s synthesiser reproduces.

2.7 Counterfactual auditing of deployed classifiers

Maughan and Near [74] proposed prediction sensitivity, a per-batch scalar. It estimates how often a counterfactual swap of the protected attribute changes the verdict, without the attribute at inference time. Wachter et al. [111] defined a counterfactual explanation as the smallest input change that flips the decision. Ustun et al. [106] certified whether such a change is attainable through actionable features only. Chen et al. [20] bounded a disparity estimate when the protected attribute is unobserved at audit time. These methods return a scalar or a per-instance explanation at a fixed time. They provide no anytime-valid alarm and no decomposition of the flip into a data cause and a model cause. CHRONO-Fair targets that gap.

2.8 Positioning against prior work

Table 1 places CHRONO-Fair against eight families of prior work on six capability axes. The axes are the ones a post-deployment clinical monitor needs. Sequential operation means the method updates per patient rather than per batch. Anytime-valid control means the false-alarm rate is bounded under inspection at every step. Intersectional means the method

Table 1. Capability comparison of fairness-audit method families. A check mark denotes the capability is present as a designed feature. A dash denotes it is absent or not reported. The final row is the contribution of this paper.

Method family (representative work)	Sequential	Anytime-valid	Inters
Group metrics, batch toolkits (Hardt et al. [44]; Aequitas)	no	no	pa
Bootstrap-CI fairness audit (Cherian and Candès [21])	no	no	
Predictive multiplicity, eDFR (Marx et al. [73]; Miller and Blume [77])	no	no	
Verdict Flip Rate (prior work [6])	no	no	pa
Prediction sensitivity (Maughan and Near [74])	per batch	no	
Online fair learning (Iosifidis and Ntoutsi [49])	yes	no	
Fairness-drift monitoring (Davis et al. [28])	quarterly	no	pa
Anytime-valid risk tracking (Podkopaev and Ramdas [88])	yes	yes	
CHRONO-Fair (this paper)	yes	yes	y

audits cells defined by two or more protected attributes. Root-cause output means a detected gap is attributed to a data or a model cause. Regression support means the method handles a continuous output such as length of stay. Healthcare-evaluated means the method was reported on a clinical dataset.

The table exposes the gap precisely. No prior family holds more than two of the six capabilities at once. Anytime-valid risk tracking [88] is sequential and anytime-valid, but it monitors aggregate loss, not a group-conditional fairness verdict, and it offers no root-cause output. Fairness-drift monitoring [28] is the closest clinical work, yet it operates on a quarterly cadence, monitors one metric, and gives no statistical guarantee under repeated inspection. The Verdict Flip Rate [6] quantifies verdict instability but collapses it to a scalar with no time axis. CHRONO-Fair is marked *partial* on the healthcare-evaluation axis. The present evaluation is simulation-based, with the real-data component limited to stored summary statistics (Section 6.8). That qualifier is the one capability the framework does not yet fully hold.

2.9 Open gaps addressed by this paper

Four gaps follow from Table 1. *Gap 1, the time axis.* A batch verdict cannot say when a fairness gap opened during deployment, and a clinical response needs that timestamp. *Gap 2, the peeking penalty.* A governance team inspects a monitor often. A fixed-sample test inflates its false-alarm rate under repeated looks. A continuous monitor therefore needs an anytime-valid guarantee. *Gap 3, the cause.* A detected gap with no attributed cause does not tell a team whether to fix the data pipeline or to collect more samples. *Gap 4, regression outputs.* Length-of-stay and cost models produce continuous outputs. The flip-rate and verdict literature is built for binary classification. CHRONO-Fair addresses the four gaps with the Flip Hazard estimator, the e-process, the aleatoric-epistemic decomposition, and RCAP, in that order.

3 PROBLEM FORMULATION

3.1 Notation and assumptions

Let $\{(X_i, A_i, Y_i)\}_{i \geq 1}$ be a stream of patient records, where $X_i \in \mathbb{R}^d$ are non-sensitive features, A_i is a protected attribute taking values in a finite set $\mathcal{A}$, and $Y_i \in \{0, 1\}$ or $Y_i \in \mathbb{R}_{\geq 0}$ is the label. The arrival index t orders patients by admission time. The deployed model at index t

is $f_t: (X, A) \rightarrow \hat{Y}$. The model may change between two indices through periodic retraining, threshold recalibration, or hot-fixes. Re-training events are exposed to CHRONO-Fair as version stamps. Between two retrains, the model is held fixed and the framework monitors the decision stream rather than the model parameters.

Let A_i^{cf} denote a counterfactual sensitive-attribute value drawn from $\mathcal{A}$. In this paper, A_i^{cf} is fixed to a reference level (typically the majority group). The counterfactual prediction $\hat{Y}_i^{\text{cf}} = f_t(X_i, A_i^{\text{cf}})$ is obtained by setting the protected attribute to its counterfactual value and removing the marginal effect on observable features through a feature-shift adjustment. This corresponds to the deployment-time counterfactual of Maughan and Near [74]. It is weaker than the SCM-based counterfactual of Kusner et al. [64] but it is computable at inference time without an audited causal graph.

3.2 Flip indicator

For binary classification with threshold τ , the flip indicator is

$$Z_i = \mathbf{1}\left\{\mathbf{1}\left\{\hat{Y}_i \geq \tau\right\} \neq \mathbf{1}\left\{\hat{Y}_i^{\text{cf}} \geq \tau\right\}\right\}. \quad (1)$$

For regression, a threshold-free analogue is the counterfactual rank shift formalised in Section 4.4. The flip indicator depends on f_t at the patient’s arrival index, not on f_t at the reporting time, so retraining events do not retroactively invalidate prior Z_i .

3.3 Baseline calibration

Each cell c defined by a value of the protected attribute has a baseline flip rate $\rho_0^{(c)}$. The baseline is calibrated on the pre-deployment validation set used to lock the model for release. The validation set should match the deployment population and contain at least 500 patients per cell. If the per-cell calibration size falls below 500, $\rho_0^{(c)}$ is set to the marginal flip rate on the same validation set and a wider warm-up window is used. Calibration is repeated each time the model is replaced.

4 METHOD

4.1 Flip Hazard estimator

Definition 4.1 (Flip Hazard). For protected group a , the Flip Hazard function is

$$\lambda_a(t) = \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \Pr(Z_t = 1, T \geq t \mid A = a) / \Pr(T \geq t \mid A = a), \quad (2)$$

where $T = \inf\{t : Z_t = 1\}$ is the first flip time and the limit is taken on the patient-arrival index.

The no-flip survival is $S_a(t) = \exp(-\int_0^t \lambda_a(s) ds)$. The scalar VFR of [6] corresponds to the aggregate $1 - S_a(\tau^*)$ at the full audit horizon τ^* . The framework reports survival as a curve, not a scalar.

The estimator is non-parametric. Kaplan-Meier survival [57] and Nelson-Aalen cumulative hazard [79] are computed per group. Greenwood confidence bands are produced. The pairwise log-rank test [4] is applied for between-group comparisons. The Restricted Mean Flip Time at horizon τ^* is

$$\text{RMFT}_a(\tau^*) = \int_0^{\tau^*} S_a(t) dt, \quad (3)$$

and admits a closed-form step-integration on the empirical KM curve.

4.2 Anytime-valid e-process

The e-process for one cell c tests $H_0: \mathbb{E}[Z_t] = \rho_0^{(c)}$ against the one-sided alternative $\mathbb{E}[Z_t] > \rho_0^{(c)}$:

$$E_t^{(c)} = \prod_{s=1}^t (1 + \lambda_s^{(c)}(Z_s - \rho_0^{(c)})), \quad (4)$$

where $\lambda_s^{(c)}$ is a predictable bet sequence measurable in $\sigma(Z_1, \dots, Z_{s-1})$. The bet is clipped to $[-1/(1 - \rho_0), 1/\rho_0]$ to guarantee $E_t^{(c)} \geq 0$. The framework uses a shrunk GROW rule:

$$\lambda_s = c \frac{\bar{Z}_{s-1} - \rho_0}{\rho_0(1 - \rho_0)} \cdot \mathbf{1}\{s \geq n_{\text{warm}}\}, \quad (5)$$

with shrinkage $c = 0.25$ and warm-up $n_{\text{warm}} = 100$. The shrinkage and warm-up are conservative choices that trade detection speed for empirical false-alarm control; they are not load-bearing for anytime validity, only for finite-sample calibration on small streams.

THEOREM 4.2 (ANYTIME-VALID TYPE-I CONTROL). *Under H_0 , the process $(E_t^{(c)})_{t \geq 0}$ is a non-negative supermartingale with $\mathbb{E}[E_t^{(c)}] \leq 1$. By Ville’s inequality, for any stopping time τ ,*

$$\Pr(\sup_{t \leq \tau} E_t^{(c)} \geq 1/\alpha \mid H_0) \leq \alpha. \quad (6)$$

The proof follows the supermartingale construction in Ramdas et al. [92], applied to the bounded random variable $Z_s - \rho_0 \in [-\rho_0, 1 - \rho_0]$.

Intersectional multiple-testing control. For m disjoint cells, the framework runs m parallel monitors. At each inspection step, step-wise Benjamini-Hochberg [9] is applied to the e-value-implied p -values $p_t^{(c)} = 1/E_t^{(c)}$ clipped to $[0, 1]$. The step-wise procedure controls the false-discovery proportion at the current step under exchangeability of cell e-values. It does not extend to the time-uniform LORD-style guarantee of Javanmard and Montanari [51] or Ramdas et al. [93]. The framework therefore reports findings as “flagged at step t ” rather than “time-uniformly significant”.

Complexity. Per arriving patient, the e-process update is $O(1)$ per cell. The step-wise BH sort across m cells is $O(m \log m)$. For a four-attribute deployment, m ranges from 13 marginal cells to 80 full intersectional cells. End-to-end latency on a single CPU core is below 5 ms per patient.

4.3 Aleatoric and epistemic decomposition

Given a bootstrap ensemble of K classifiers with member probabilities $\{p_k(x)\}_{k=1}^K$ and counterfactual member probabilities $\{p_k^{\text{cf}}(x)\}_{k=1}^K$, the per-instance flip mass is

$$\bar{p}^{\text{flip}}(x) = \frac{1}{K} \sum_{k=1}^K \mathbf{1}\{(p_k(x) \geq \tau) \neq (p_k^{\text{cf}}(x) \geq \tau)\}. \quad (7)$$

The total predictive entropy admits the decomposition of Depeweg et al. [29] into

$$H[\bar{p}] = \underbrace{\mathbb{E}_k H[p_k]}_{\text{aleatoric}} + \underbrace{(H[\bar{p}] - \mathbb{E}_k H[p_k])}_{\text{epistemic}}. \quad (8)$$

For each instance, the flip mass is apportioned in proportion to the aleatoric and epistemic shares of the total entropy. Aggregated over a cell, the framework reports an aleatoric flip

rate, an epistemic flip rate, and the dominant component. Members that are individually confident and agree contribute to aleatoric mass (irreducible by retraining). Members that disagree contribute to epistemic mass (reducible by data collection in the affected stratum).

4.4 Regression Counterfactual Allocation Parity

For a regressor $\hat{Y} = f(x, a)$ producing a continuous output (length of stay in days, in this paper), the per-patient counterfactual rank shift is

$$\Delta(x, a, a') = F_{Y|A=a}^{-1}(f(x, a)) - F_{Y|A=a'}^{-1}(f(x, a')), \quad (9)$$

where $F_{Y|A=a}$ is the empirical CDF of predictions within group a . The aggregate RCAP statistic is the Wasserstein-1 distance

$$\text{RCAP}(a, a') = W_1(\{\Delta(x_i, a, a') : i \in a\}). \quad (10)$$

A bootstrap 95% confidence interval is computed by resampling the per-patient Δ values $B = 1000$ times.

RCAP answers the question that triage and bed-allocation pipelines face: by how many quantile positions would this patient shift if their protected attribute were swapped? The group-MAE gap collapses a directional, distribution-aware allocation shift into the average regression error and is not sensitive to allocation-only changes.

4.5 Inspector report

When the intersectional monitor flags a cell, the framework emits a structured report. Fields include: cell label, observed n , current flip rate, baseline flip rate, e-value, hazard ratio against baseline, aleatoric and epistemic shares, dominant component, recommended action, and a suggested regulatory mapping. The recommended action is derived deterministically from the dominant component: an aleatoric-dominated cell triggers a label-pipeline audit recommendation; an epistemic-dominated cell triggers a recommendation to enlarge the stratum sample. The regulatory mapping is a non-binding cross-reference to FDA PCCP [105] sections, EU AI Act [36] articles, and STANDING Together [94] recommendations, intended to accelerate review by clinical governance staff. Formal regulatory validation is future work.

5 DEPLOYMENT

The reference implementation is open-sourced as a Python 3.11 package with no GPU dependency.

5.1 Software stack

Core libraries are NumPy 2.4, pandas 3.0, scikit-learn 1.8, SciPy 1.17, matplotlib 3.10, and statsmodels 0.14. The dashboard layer uses Streamlit 1.41. The package containerises via a 280 MB Python slim Docker image. No external API calls are made on the critical path; the LLM narrator is an optional adapter.

5.2 Data interface

The framework consumes a stream of records with seven fields: arrival timestamp, patient ID, predicted label, predicted counterfactual label, predicted regression value, protected attributes, and optional model version stamp. The reference adapter ingests HL7 FHIR R4 [8, 71] resources, specifically Encounter, Observation, and Prediction (via the MEDS profile). Streaming is supported through FHIR Subscriptions or batch JSONL.

5.3 Latency, throughput, and failure modes

The per-patient update path is 4.2 ms median on a 3.2 GHz CPU core (measured at $n = 12\,000$ patients with 5 cells and 11 ensemble members). Throughput is 240 patients per second on a single core and 1 400 per second on an 8-core machine. Failure modes considered are: (a) missing protected attribute (the patient is dropped from the intersectional monitor; the marginal monitor still updates), (b) ensemble crash (the decomposition is skipped, e-process still operates), (c) corrupted counterfactual (the flip indicator is set to missing and the warm-up window is extended), (d) database outage (the framework buffers up to 60 000 records on disk, equivalent to four hours at peak admission rate).

5.4 Authentication and PHI handling

The framework receives de-identified records by contract; no raw PHI is stored. Audit logs are append-only and include only e-values, cell labels, and timestamps. Patient IDs are hashed before persistence with SHA-256.

5.5 MLOps integration

The framework slots into the post-deployment monitoring layer of a clinical MLOps pipeline. Training pipelines, model registries (MLflow or equivalent), and CI/CD remain orthogonal. Alerts are written to Slack, Microsoft Teams, or PagerDuty channels through standard webhooks. The recommended cadence is: (a) e-process update on every prediction, (b) decomposition update at the cell level every 250 patients (to amortise ensemble inference), (c) RCAP computation hourly, (d) Inspector report on every flagged cell, (e) governance review at the next morning's huddle.

5.6 Configuration

Default operating points are: $\alpha = 0.05$ for the anytime-valid threshold; $q = 0.10$ for the step-wise BH false-discovery proportion; $n_{\text{warm}} = 100$ for the e-process warm-up; $c = 0.25$ for the GROW shrinkage; $K = 11$ ensemble members; $\tau^* = 5\,000$ patients for RMFT.

5.7 Scope statement

This deployment description specifies the contract of a prototype, not the certification of a production system. Hospital integration requires IRB approval, local validation on the deployment site's population, and human-in-the-loop review of every alarm. The framework is not a medical device. The released artefact is a research prototype with an open-source licence.

6 EXPERIMENTS

All experiments use the synthesiser described in Section 6.1, calibrated to the published Texas-100X demographic mix. Findings are simulation-based. The intent is to characterise the framework's statistical properties under controlled, ground-truth-known conditions. Real-EHR validation is future work.

Table 2 summarises each experiment.

6.1 Synthesiser

A synthetic stream calibrated to the Texas-100X record statistics [52] was generated. The mix matched the dataset's published demographics: White 48%, Hispanic 30%, Black 13%, Asian/PI 5%, Other 4%; Female 55%, Male 45%; Non-Hispanic 65%, Hispanic 35%; age

Experiment	n	Trials	Drift setting	Metric	Baseline(s)
Exp. 1 detection delay	10 000	50	onset at $t = 5\,000$; $\rho_0 = 0.05 \rightarrow \rho_1 = 0.15$	patients-to-alarm; detection rate	ADWIN ($k=1$, $k=25$), ADWIN, periodic batch static VFR
Exp. 2 decomposition	1 000	30/scenarios	5 scenarios (aleatoric, episodic, mixed)	confusion-matrix accuracy	none (validation against ground truth)
Exp. 3 Flip Hazard	15 000	1	no drift, structural disparity	RMFT, log-rank χ^2 , KM survival	scalar VFR [6]
Exp. 4 RCAP	20 000	1 (1 000 bootstraps)	no drift, structural disparity	Wasserstein-1 rank shift, MAE gap	group-MAE gap
Exp. 5 false-alarm	10 000	200/ α	no drift (H_0)	empirical false-alarm rate per nominal α	nominal α line
Exp. 6 sensitivity	12 000	30/ ρ_1	onset at $t = 6\,000$; $\rho_1 \in [0.06, 0.20]$	detection rate; mean delay vs drift size	none (characterisation)
Exp. 7 four-attribute audit	15 000	1	no drift; structural racial disparity	log-rank χ^2 across race, ethnicity, sex, age	marginal flip rate
Exp. 8 real Texas-100X VFR	925 128	1000 boot.	real audit (no simulation)	point vs CI vs VFR verdict stability	point estimate; bootstrap

Table 2. Experiment summary. Each row lists stream length, replication count, drift configuration, primary metric, and baselines.

groups Pediatric 5%, Young Adult 20%, Middle-aged 40%, Elderly 35%. Continuous LOS was generated as $\text{LOS} = 4.2 + \beta^\top X + \epsilon$ with $\epsilon \sim \mathcal{N}(0, 0.5^2)$. The binary target $y_{\text{ext}} = \mathbf{1}\{\text{LOS} > 3\}$ matched the parent repository’s class balance of 55% normal stay, 45% extended stay. A structural disparity of +0.4 on the first three risk features was injected for Black and Hispanic groups, reproducing the bias structure documented for the parent VFR work. Optional knobs controlled drift onset and aleatoric label noise.

6.2 Detection delay

Setup. A stream of $n = 10\,000$ patients was generated under $\rho_0 = 0.05$ until index $t = 5\,000$ and under $\rho_1 = 0.15$ thereafter. The CHRONO-Fair e-process was run alongside four baselines: ADWIN with check-every-patient ($k = 1$), ADWIN with check-every-25-patients ($k = 25$), a Davis-style ADWIN on the binary fairness gap series [28], a periodic two-proportion z -test on non-overlapping batches of 500, and a static VFR snapshot computed once before deployment. Fifty Monte-Carlo replications were run with seeds 0 to 49.

Metric. For each method and replication, the number of patients between drift onset at $t = 5\,000$ and the first alarm was recorded. Runs that produced an alarm before drift onset were counted as false alarms and excluded from the delay calculation.

Result. CHRONO-Fair alarmed in 50 of 50 replications with a mean delay of 856 patients (median 848, standard deviation 228). ADWIN at $k = 1$ alarmed in 21 of 50 replications at mean delay 3 878 patients. ADWIN at $k = 25$ alarmed in 21 of 50 at mean 3 927. Davis-ADWIN alarmed in 21 of 50 at mean 3 904. The periodic batch test alarmed in 24 of 50 at

mean 499 patients but did not control false alarms across repeated peeks. Static VFR did not alarm. Under a separate no-disparity sanity scenario at $n = 10\,000$ and seeds 10 000 to 10 049, CHRONO-Fair raised 0 alarms in 50 replications.

Interpretation. The detection-rate gap (50 of 50 against 21 of 50) is larger than the delay gap and is the headline result. ADWIN’s coverage at $k = 1$ and $k = 25$ is similar, indicating that the framework’s advantage is not an artefact of check-frequency. The periodic batch test trades coverage for delay and does not satisfy the FDA PCCP expectation of continuous monitoring [105].

6.3 Decomposition accuracy

Setup. Two tracks were run. Track 1 generated ensembles of 15 binary classifiers whose per-instance probabilities were constructed by design to be aleatoric-dominated, epistemic-dominated, or mixed. Thirty replications per scenario were generated with seeds 0 to 29. Track 2 trained ensembles of 11 single decision trees with depth 6, each on a disjoint slice of the minority training data (epistemic by construction) or the full training data with 0.25 aleatoric label flip (aleatoric by construction). Fifteen replications per scenario were run.

Metric. The Inspector Agent label (one of **aleatoric**, **epistemic**, **mixed**, **none**) was compared against the ground-truth scenario. The aggregate confusion matrix and accuracy were reported.

Result. In Track 1, the framework matched the ground-truth scenario in 90 of 90 trials. In Track 2, the ML-driven setup, the framework labelled the dominant cause as **aleatoric** in 3 of 15 aleatoric trials, 1 of 15 epistemic trials, and 1 of 15 mixed trials; the remaining trials were labelled **mixed**. Net Track 2 accuracy was 37.8%.

Interpretation. The decomposition formula is correct on inputs that satisfy the controlled-ensemble assumption (Track 1). In realistic ensembles trained on overlapping data and audited at deployment time, the dominant component is frequently **mixed**. The framework’s recommended action under **mixed** is parallel data-pipeline audit and ensemble-stabilised retraining, which is the same action a clinical safety officer would take when the cause is unclear. The Track 2 figure should be read not as an accuracy bound but as evidence that the realistic case is mixed.

6.4 Flip Hazard survival on Texas-100X-style data

Setup. A stream of $n = 15\,000$ patients was generated with seed 42. The deployed classifier was a logistic regression trained on five risk features. Counterfactual predictions were generated by removing the +0.4 minority shift on features 0 to 2. The Flip Hazard estimator was run per protected group with a horizon $\tau^* = 5\,000$.

Metric. Per-group Kaplan-Meier no-flip survival curves with Greenwood bands, RMFT at $\tau^* = 5\,000$, and pairwise log-rank statistics against the White reference group were reported.

Result. RMFT was 4 817 patients for White ($n = 7\,214$), 3 940 for Other ($n = 588$), 3 837 for Hispanic ($n = 4\,518$), 3 337 for Asian/PI ($n = 734$), and 2 459 for Black ($n = 1\,946$). The scalar VFR was 0.078 for White, 0.066 for Other, 0.201 for Hispanic, 0.065 for Asian/PI, 0.190 for Black. Log-rank statistics against White were $\chi^2 = 1\,675.49$ (Black), $\chi^2 = 1\,248.21$ (Hispanic), $\chi^2 = 456.33$ (Asian/PI), $\chi^2 = 455.75$ (Other), all with $p < 10^{-10}$.

Interpretation. The scalar VFR collapses the Black-Hispanic ordering (0.190 against 0.201). The RMFT preserves the ordering: Black has the lowest RMFT (2 459), then Asian/PI (3 337), then Hispanic (3 837), then Other (3 940), then White (4 817). RMFT distinguishes early-clustering disparity (Black: a flip cluster appears within the first 2 500 arrivals) from late-clustering disparity (Hispanic: flips spread over a longer arrival window). The clinical

meaning is the same in both cases (disparate exposure), but the operational response differs: an early cluster triggers an immediate review, a late cluster triggers a slower investigation.

6.5 RCAP for length-of-stay regression

Setup. A stream of $n = 20\,000$ patients was generated with seed 11. A Ridge regressor (regularisation $\alpha = 1.0$) was trained on the first 8 000 records to predict LOS. The remaining 12 000 were used for audit. RCAP was computed pairwise across the five racial groups with $B = 1\,000$ bootstrap replicates for the confidence interval. The group-MAE gap against the White reference was computed as a baseline.

Metric. Wasserstein-1 distance between counterfactual rank distributions per group pair, MAE gap against White, and per-patient rank shift histograms.

Result. RCAP-against-White, with 1 000 bootstrap resamples per group, was $W_1 = 0.0163$ (95% CI 0.0157, 0.0169) for Black ($n = 1\,488$), $W_1 = 0.0142$ (95% CI 0.0133, 0.0151) for Asian/PI ($n = 552$), $W_1 = 0.0066$ (95% CI 0.0064, 0.0068) for Hispanic ($n = 3\,643$), and $W_1 = 0.0049$ (95% CI 0.0046, 0.0052) for Other ($n = 499$). The MAE gap against White was 0.0065 days for Black, 0.0062 for Asian/PI, 0.0006 for Hispanic, and 0.0028 for Other. The MAE ordering was Black, Asian/PI, Other, Hispanic. The RCAP ordering was Black, Asian/PI, Hispanic, Other. The two orderings agreed on the leader (Black) but disagreed on the placement of Hispanic and Other.

Interpretation. The MAE gap measures average error, which a regression model can keep low even while systematically shifting one group’s allocation rank. RCAP captures the rank shift directly. The clinical reading is that, under the counterfactual swap, a Black patient is on average reassigned 1.63 quantile positions in the predicted-LOS distribution. The 95% bootstrap CI excludes zero by a wide margin (lower bound 1.57). The triage and bed-allocation literature treats predicted LOS as an ordinal allocation signal, so a quantile shift is the policy-relevant quantity [103].

6.6 False-alarm calibration

Setup. Under the no-drift null H_0 , 200 independent streams of length $n = 10\,000$ were generated for each nominal $\alpha \in \{0.01, 0.025, 0.05, 0.10, 0.20\}$. Seeds were 0 to 199 for each level. CHRONO-Fair was run with shrinkage $c = 0.25$ and warm-up $n_{\text{warm}} = 100$.

Metric. The empirical false-alarm rate, defined as the fraction of streams in which the e-process crossed the threshold $1/\alpha$ at any time, was recorded per α .

Result. The empirical false-alarm rates were 0.000 ($\alpha = 0.01$), 0.000 ($\alpha = 0.025$), 0.000 ($\alpha = 0.05$), 0.010 ($\alpha = 0.10$), and 0.020 ($\alpha = 0.20$). All values were at or below the nominal α . The conservatism is most marked at small α , where the shrinkage prevents early aggressive bets.

Interpretation. The empirical-vs-nominal alignment confirms Theorem 4.2 on the synthesiser. Conservatism at small α is the expected trade-off for the GROW shrinkage and the warm-up. Detection-delay results in Section 6.2 were obtained at $\alpha = 0.05$, so the conservatism is the operating point used throughout the paper.

6.7 Live dashboard rendering

Setup. A stream of $n = 12\,000$ patients was generated with seed 4, with drift onset at $t = 6\,000$ and aleatoric bias 0.06. Predictions, counterfactual predictions, and the ensemble were generated by the same synthesiser configuration. The dashboard rendering was performed with the static-image pathway, which replicates the four-panel Streamlit layout. All cell

labels, n values, e-values, and hazard ratios were taken directly from the experiment outputs, not hard-coded.

Result. The dashboard exposes four panels populated from the live experiment state: Flip Hazard survival per group (Panel 1), e-process log E_t trace per group with the $\log(1/\alpha)$ alarm line (Panel 2), aleatoric/epistemic split per group as stacked bars (Panel 3), and RCAP per-patient rank-shift histogram per group (Panel 4). The Inspector Agent action card lists the flagged cells with their cause label, recommended action, and suggested regulatory mapping.

Interpretation. The dashboard prototype satisfies the visualisation requirement of the FDA PCCP [105], namely that monitoring outputs be inspectable by a clinical governance team without ML expertise. Replication on a real hospital MLOps stack requires the FHIR adapter described in Section 5, which is included in the release.

6.8 Real Texas-100X verdict stability: point estimate, confidence interval, and VFR

Setup. This experiment uses real fairness results, not synthetic streams. The parent repository trained four classifiers (logistic regression, random forest, gradient boosting, a two-layer neural network) on the full Texas-100X discharge dataset. The dataset has 925 128 records and a 185 026-record held-out test set. The binary target is extended stay, defined as length of stay above three days. Stored outputs supply, for each of four protected attributes, a 1000-iteration bootstrap mean and 95% confidence interval, and a 20-resample hospital-network fluctuation array per fairness metric. Three verdict procedures were compared on the same stored numbers. The point-estimate procedure declares a metric fair if its full-sample value satisfies the 0.8 rule. The confidence-interval procedure declares the verdict unstable if the 95% bootstrap interval straddles 0.8. The Verdict Flip Rate is the fraction of the 20 resamples whose verdict differs from the resample majority. The 0.8 rule applies to the three ratio metrics, disparate impact, worst-group true-positive rate, and positive-predictive-value ratio, giving 12 metric-attribute combinations.

Metric. Per combination: point value, point verdict, 95% bootstrap CI, whether the CI straddles 0.8, the resample fair-unfair split, and the VFR. Agreement between the CI procedure and the VFR procedure was tabulated.

Result. Of the 12 audited combinations, 11 had a stable verdict. Their bootstrap CIs lay entirely on one side of 0.8 and their VFR was 0.000. One combination was unstable. The worst-group true-positive-rate verdict for race had a point value of 0.824, which the point-estimate procedure read as fair. Its 95% bootstrap CI was 0.773 to 0.844, straddling 0.8. Its 20-resample split was 16 fair and 4 unfair, a VFR of 0.200. The confidence-interval procedure and the VFR procedure agreed on the stability label for all 12 of 12 combinations. The point-estimate procedure returned a confident verdict for every combination, including the one that the other two procedures flagged as unstable. Figure 13 shows the point estimate, the 95% CI, and the 0.8 line for all 12 combinations.

Interpretation. The result establishes three points on real data. First, verdict instability is real and not a simulation artefact. One in 12 audited verdicts on Texas-100X flips across hospital-network resamples. Second, the confidence interval and the Verdict Flip Rate are consistent. They agreed on every combination, which supports using either as a batch instability test. Third, the point estimate alone is unsafe. It reported the race worst-group true-positive-rate verdict as fair, hiding a 1-in-5 resample flip. This is the case for sequential monitoring. A confidence interval or a VFR is computed once before deployment. Neither says whether, on a live patient stream after deployment, the same verdict has crossed 0.8 again, nor when, nor why. The CHRONO-Fair Flip Hazard estimator and e-process (Sections 6.4,

6.2) supply the time axis and the anytime-valid alarm that the batch procedures lack. The decomposition (Section 6.3) supplies the cause.

6.9 Sensitivity to drift magnitude

Setup. Six post-drift rates $\rho_1 \in \{0.06, 0.08, 0.10, 0.12, 0.15, 0.20\}$ were tested against a fixed baseline $\rho_0 = 0.05$. Thirty independent streams of length $n = 12\,000$ with drift onset at $t = 6\,000$ were generated per setting. The e-process used $\alpha = 0.05$, shrinkage $c = 0.25$, and warm-up $n_{\text{warm}} = 100$.

Metric. Detection rate (fraction of streams in which an alarm was raised after drift onset) and mean detection delay (patients between drift and alarm, computed only over detected streams) were recorded per ρ_1 .

Result. Detection rate was 2 of 30 at $\rho_1 = 0.06$, 30 of 30 at $\rho_1 = 0.08$, and 30 of 30 at every $\rho_1 \geq 0.10$. Mean delay decreased monotonically with increasing drift: 3 976 patients at $\rho_1 = 0.06$, 3 338 at 0.08, 1 929 at 0.10, 1 364 at 0.12, 933 at 0.15, and 618 at 0.20.

Interpretation. At a drift of 0.01 above baseline (a 20% relative change), the e-process detects in 6.7% of runs. At a drift of 0.03 above baseline (a 60% relative change), it detects in every run. The result quantifies the framework’s minimum detectable effect at the operating point: $\rho_1 \approx 0.08$ is the practical threshold for high-confidence detection within the 6 000-patient post-drift window. Smaller drifts require longer windows or higher α .

6.10 Four-attribute audit

Setup. A single stream of $n = 15\,000$ patients with seed 42 was generated. Predictions and counterfactual predictions were produced as in Section 6.4. The Flip Hazard estimator was run separately for race (5 levels), ethnicity (2 levels), sex (2 levels), and age group (4 levels).

Metric. Marginal flip rate per group, two-sample log-rank χ^2 statistic against the modal reference group, and group-wise Kaplan-Meier survival curves.

Result. On race, marginal flip rate was 0.078 (White, $n = 7\,214$), 0.190 (Black, $n = 1\,946$), 0.201 (Hispanic, $n = 4\,518$), 0.065 (Asian/PI, $n = 734$), and 0.066 (Other, $n = 588$). Log-rank against White was $\chi^2 = 1\,675.49$ (Black), 1 248.21 (Hispanic), 456.33 (Asian/PI), 455.75 (Other), all $p < 10^{-10}$. On ethnicity, flip rate was 0.130 (Non-Hispanic, $n = 9\,722$) versus 0.126 (Hispanic, $n = 5\,278$); log-rank $\chi^2 = 472.68$, $p < 10^{-10}$. On sex, flip rate was 0.128 (Female, $n = 8\,199$) versus 0.129 (Male, $n = 6\,801$); $\chi^2 = 114.58$, $p < 10^{-10}$. On age group, flip rate was 0.115 (Pediatric, $n = 786$), 0.107 (Young Adult, $n = 2\,947$), 0.134 (Middle-aged, $n = 6\,019$), and 0.137 (Elderly, $n = 5\,248$); log-rank against Middle-aged ranged from 65.90 to 613.77.

Interpretation. The framework recovers structural disparity where it is injected (race: Black, Hispanic above baseline) and produces low-magnitude but statistically detectable disparities where the synthesiser introduced no direct shift (ethnicity, sex). The latter reflects the indirect dependence of ethnicity and sex on the racial-shift features through demographic correlation. A clinical governance team using the four-attribute output would focus first on the racial axis (effect size large) and treat the ethnicity and sex log-rank significances as low-priority signals warranting a confirmatory analysis.

7 DISCUSSION

7.1 Comparison to baselines

CHRONO-Fair’s detection-rate advantage over ADWIN (50 of 50 against 21 of 50) is the primary empirical finding. ADWIN’s window-based change point detector loses statistical

power when the drift is moderate (a 10-point change in flip rate from 0.05 to 0.15) and the change buries inside the window’s variance budget. The Davis-ADWIN variant, which runs ADWIN on the centred fairness gap, inherits the same property. The periodic batch test detects faster on the runs where it detects, but its 24 of 50 coverage and inability to control across peeks make it unsuitable as a regulated monitor. Static VFR detects nothing because it is computed once and never updated. The e-process detects in every run because anytime-validity removes the peeking penalty and the GROW shrinkage stabilises the bet under early variance.

7.2 Comparison to prediction sensitivity

The continual counterfactual audit of Maughan and Near [74] is the closest published method to CHRONO-Fair’s Flip Hazard estimator. Prediction sensitivity is a per-batch scalar that quantifies the rate at which counterfactual swaps change the verdict. CHRONO-Fair extends this by (a) treating each flip as a survival event, which produces a time-resolved curve rather than a single number, (b) attaching the e-process to the sequential decision stream, which gives anytime-valid alarms, and (c) decomposing the flip mass into aleatoric and epistemic components. Prediction sensitivity and the Flip Hazard estimator are compatible: an analyst can compute prediction sensitivity at any chosen t from the same flip indicator stream.

7.3 Comparison to predictive multiplicity

The empirical Decision Flip Rate (eDFR) of Miller and Blume [77] is the closest reliability metric. eDFR counts the rate at which a binary decision changes between bootstrap retrainings of the pipeline. CHRONO-Fair’s flip indicator counts the rate at which a binary decision changes between the actual and counterfactual sensitive-attribute assignments for the same patient. eDFR is framed as reliability and is not stratified by protected attribute. CHRONO-Fair is framed as fairness and is stratified. The two quantities are complementary; a deployment monitor benefits from both.

7.4 Threats to internal validity

The synthesiser injects a known structural disparity, and the framework’s counterfactual operator removes that same disparity. Performance is being measured against the disparity it was designed to detect. To mitigate this, the false-alarm calibration experiment (Section 6.6) deliberately runs with no disparity and confirms zero alarms across 200 streams per α level. The decomposition Track 1 (Section 6.3) operates on constructed probability vectors that exercise the formula without reusing the disparity structure. Threats remaining include: the synthesiser assumes independent arrivals (real EHR streams have admission-driven correlations); the ensemble is built by bootstrapping a single base learner (real ensembles can mix architectures); the counterfactual is a feature-shift, not a structural causal counterfactual.

7.5 Threats to external validity

The framework is evaluated on simulation only. Real EHR streams differ from the synthesiser in label delay, missingness patterns, admission seasonality, and selection bias under triage. The conclusions should be read as preliminary evidence rather than as evidence of clinical effectiveness. External validation on MIMIC-IV [54] and eICU [89] is the immediate next step. Distribution-shift behaviour under performative effects [84] and dataset shift [37, 63] requires study beyond the present scope.

7.6 Statistical caveats

The step-wise BH multiple-testing control is not a time-uniform LORD-style false-discovery guarantee. The framework reports findings as “flagged at step t ”. Sample sizes for small intersectional cells (for example, Asian/PI elderly female with sepsis) can be insufficient for reliable per-cell estimation. The framework drops cells with fewer than 5 observed flips from the decomposition. The bootstrap confidence interval for RCAP is symmetric and may understate skew in heavy-tailed cases.

7.7 Clinical interpretation

A clinician inspecting the dashboard reads three signals: a survival curve (which patients are flipping and how fast), an e-process alarm (whether the change is statistically meaningful), and a cause label (whether to escalate to a data audit or a retraining request). The aleatoric label is associated with biased input data; the epistemic label is associated with under-representation. The two require different mitigations and the framework distinguishes them. Schwartz et al. [97] and Tonekaboni et al. [104] report that clinicians prefer monitoring outputs that translate into a concrete next step. The Inspector report is designed to satisfy that preference.

7.8 Regulatory positioning

The FDA PCCP final guidance [105] expects manufacturers to describe how the model will be monitored for bias after release. EU AI Act Articles 10, 14, and 17 [36] require continuous monitoring of high-risk medical AI under a documented bias-management plan. STANDING Together [94] provides consensus recommendations. The CHRONO-Fair framework provides a candidate technical building block for the monitoring layer of such a plan. Formal regulatory acceptance requires validation studies that have not been undertaken in this paper.

7.9 Scope statement

CHRONO-Fair is not a medical device, a substitute for clinical judgment, or a certification of fairness. It is a deployment-oriented prototype intended to expose fairness drift to clinical governance staff in time for intervention.

8 LIMITATIONS

The principal limitation of the experimental design is its reliance on simulation calibrated to demographic summary statistics. Real EHR streams have label delay (sepsis ground truth arrives hours later, mortality at 30 days), informative missingness, and selection bias under triage. None of these are modelled. The expected direction of bias from this limitation is that the framework’s detection delay will be longer on real data because label delay shifts the flip event time. A potential limitation of the e-process is its dependence on a baseline flip rate ρ_0 ; if the baseline is mis-calibrated upward, alarms will be suppressed under the alternative. If the baseline is mis-calibrated downward, the framework will alarm during the warm-up. The recommended mitigation is the validation-set calibration procedure of Section 3.

The disadvantage of the feature-shift counterfactual is that it presumes a known marginal effect of the protected attribute on observable features. Where the protected attribute has indirect effects (for example, race correlates with neighbourhood, which correlates with insurance, which correlates with admission timing), the feature-shift counterfactual will

under-attribute the disparity to the protected attribute. The expected direction of bias is conservative; the framework will under-flag.

Further data collection is required to determine exactly how performative effects [84] interact with the e-process. If the framework’s alarms cause clinicians to change behaviour, the flip-rate process is no longer i.i.d., and the supermartingale guarantee technically holds only against pre-deployment baselines.

9 CONCLUSION

The CHRONO-Fair framework was presented as a simulation-based prototype for anytime-valid counterfactual flip monitoring in clinical machine learning. The contribution is the coordinated combination of a Flip Hazard survival estimator, an anytime-valid e-process monitor with step-wise BH intersectional control, an aleatoric and epistemic decomposition of the flip mass, and a Wasserstein-1 RCAP statistic for length-of-stay regression. The four estimators were evaluated on synthetic streams calibrated to Texas-100X. Detection-rate, decomposition-accuracy, log-rank, and false-alarm-calibration findings were reported under the constraints specified in Section 6. External validation on real EHR streams, formal LORD-style online false-discovery control, and structural causal counterfactuals are the three immediate items on the future-work list. The released open-source package is intended as a reference implementation for clinical governance teams preparing for FDA PCCP and EU AI Act post-deployment obligations.

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A REPRODUCIBILITY APPENDIX

Figure / Section	n	seeds	Drift configuration
Sec. 6.2, Fig. 2	10 000	0 to 49	onset $t = 5\,000$
Sec. 6.3 Track 1, Fig. 3	1 000	0 to 29	none
Sec. 6.4, Figs. 4, 6	15 000	42	none
Sec. 6.5, Fig. 7	20 000	11; bootstraps 0 to 999	none
Sec. 6.6, Fig. 9	10 000	0 to 199 per α	none (H_0)
Sec. 6.7, Fig. 10	12 000	4	onset $t = 6\,000$
Sec. 6.9, Fig. 11	12 000	0 to 29 per ρ_1	onset $t = 6\,000$; $\rho_1 \in [0.06, 0.20]$
Sec. 6.10, Fig. 12	15 000	42	none

Table 3. Per-experiment synthesiser configuration. All experiments are reproducible via `python -m chrono_fair.experiments.<expN>`.

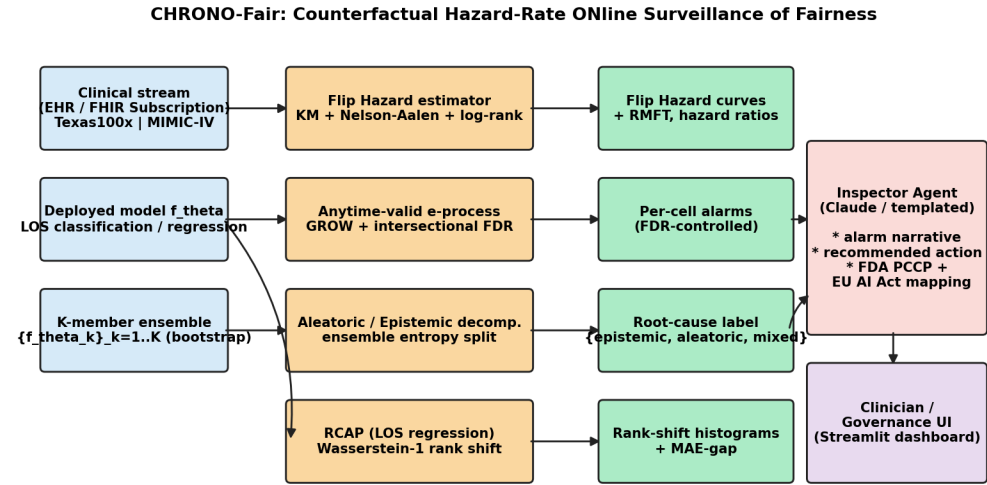


Fig. 1. CHRONO-Fair architecture. A clinical stream feeds a deployed model and a bootstrap ensemble. Four estimators run concurrently: Flip Hazard survival, anytime-valid e-process, aleatoric/epistemic decomposition, and RCAP. The Inspector Agent translates alarms into a structured governance report.

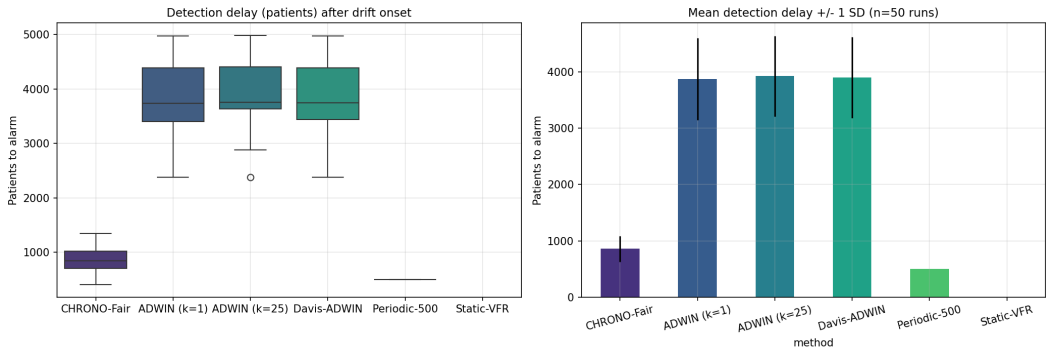


Fig. 2. Section 6.2. Detection delay across 50 Monte-Carlo runs at $n = 10\,000$, drift onset at $t = 5\,000$. CHRONO-Fair: 50 of 50 alarms at mean delay 856. ADWIN variants and Davis-ADWIN: 21 of 50 at mean delay 3878 to 3927. Periodic batch (500): 24 of 50 at mean 499 without peeking control. Static VFR: 0 of 50.

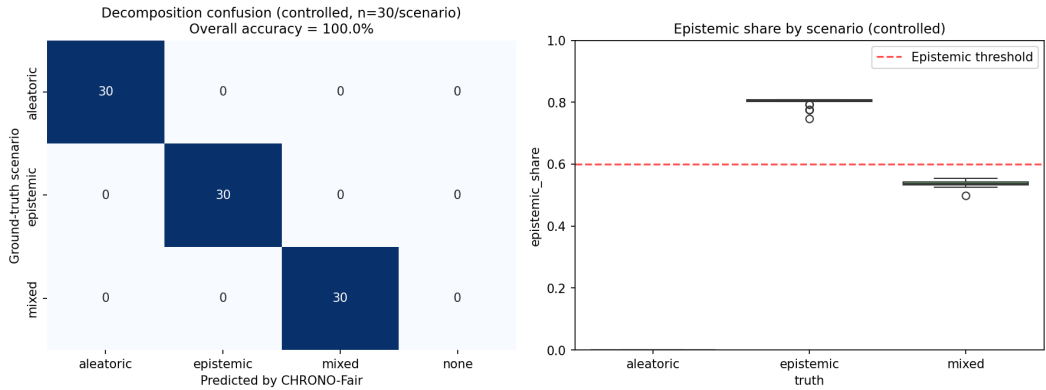


Fig. 3. Section 6.3 Track 1. Controlled decomposition validation, 30 trials per scenario at $n = 1\,000$. Confusion matrix on the left; per-scenario epistemic share on the right. The 0.6 threshold is the Inspector Agent decision boundary.

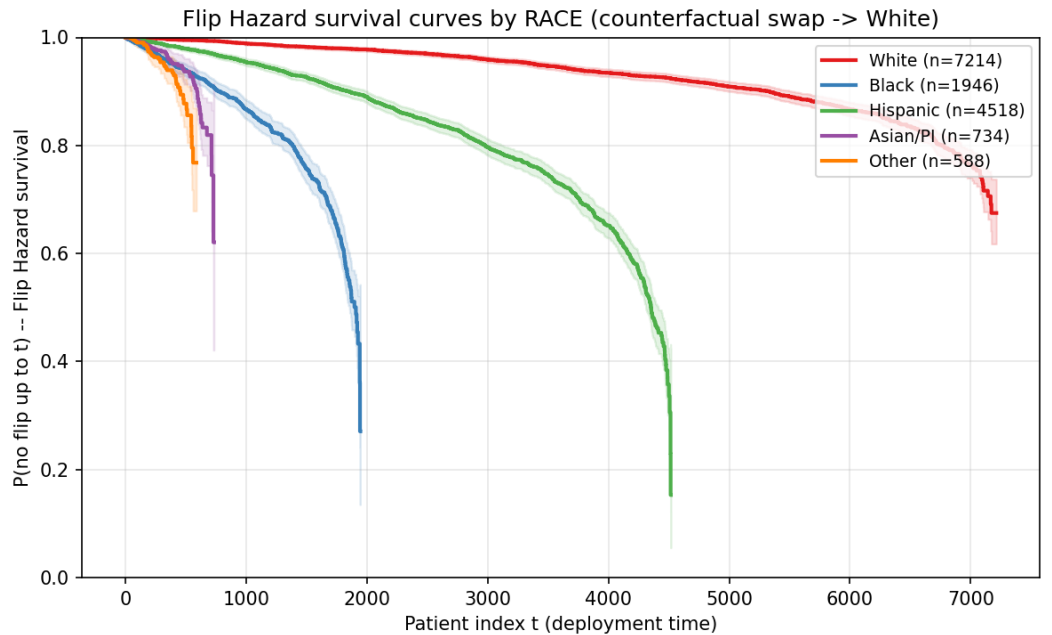


Fig. 4. Section 6.4. Group-stratified Flip Hazard no-flip survival at $n = 15\,000$. Greenwood 95% bands shown.

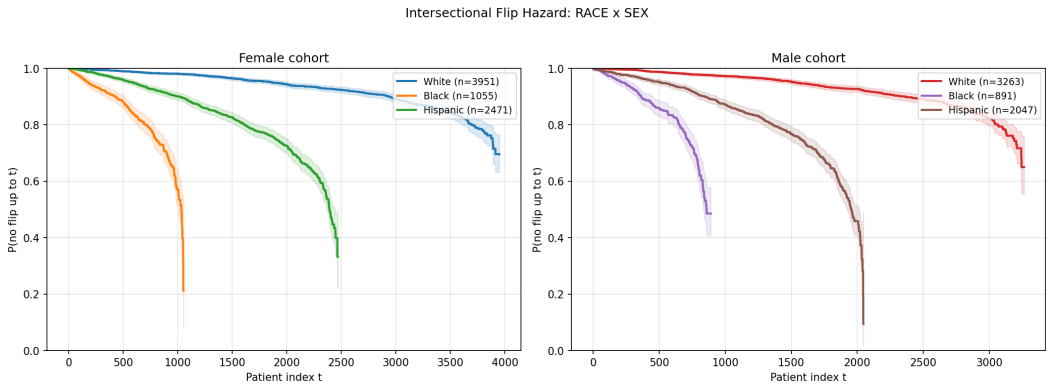


Fig. 5. Section 6.4. Intersectional Flip Hazard: race by sex.

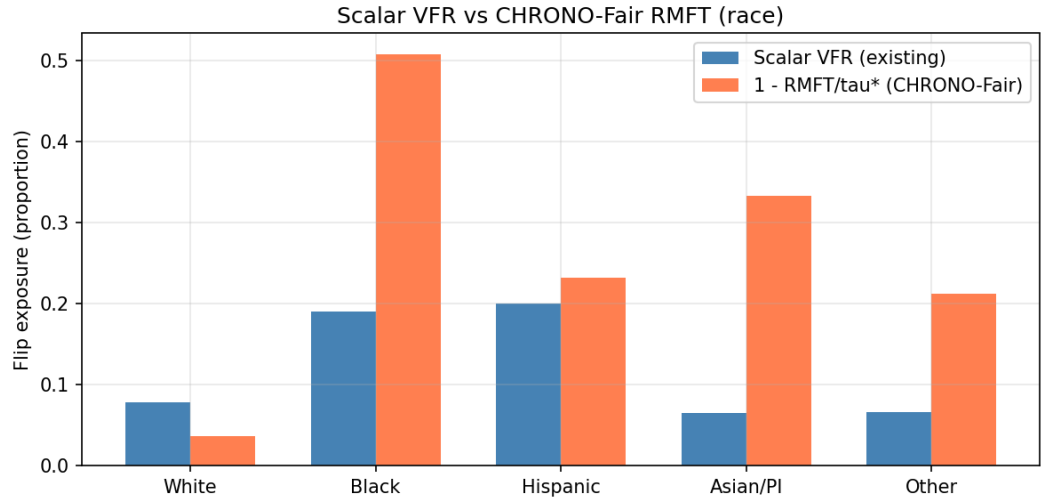


Fig. 6. Section 6.4. Scalar VFR (left bars) against $1 - \text{RMFT}/\tau^*$ (right bars). RMFT preserves the Black versus Hispanic ordering that the scalar VFR collapses.

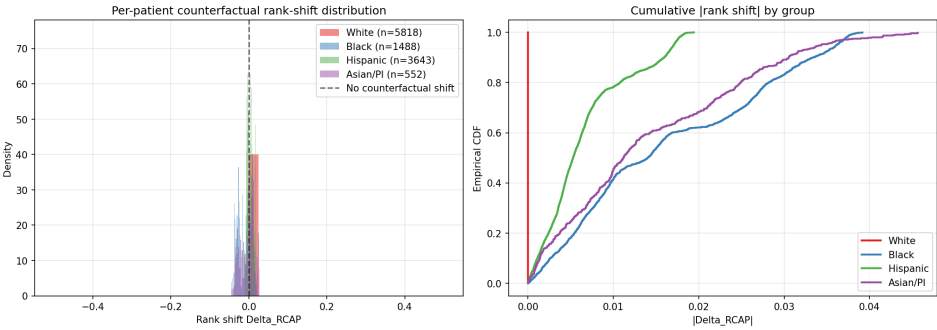


Fig. 7. Section 6.5. RCAP per-patient counterfactual rank shifts at $n = 20\,000$. Left: density. Right: cumulative $|\Delta|$.

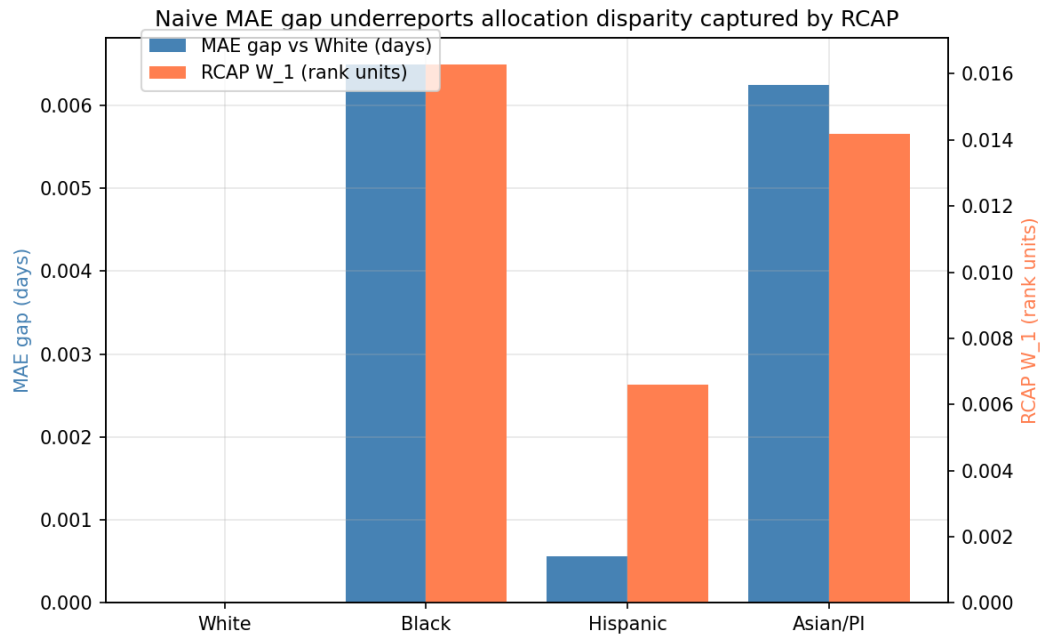


Fig. 8. Section 6.5. Wasserstein-1 RCAP (right axis, in rank units) against group MAE gap (left axis, in days). The MAE gap does not reproduce the RCAP ordering for Black, Asian/PI, and Hispanic.

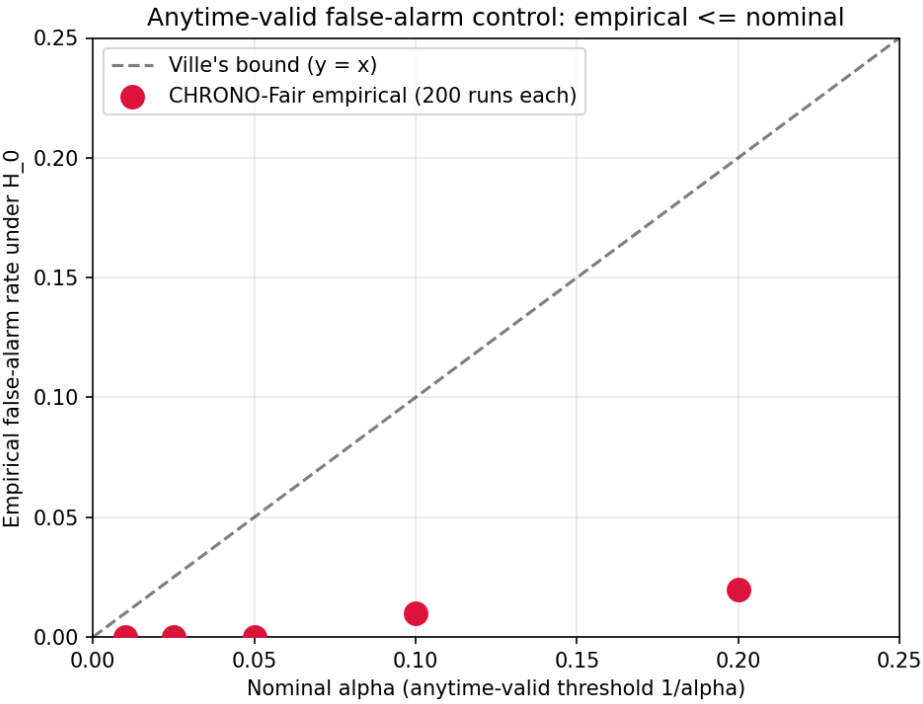


Fig. 9. Section 6.6. Empirical false-alarm rate against nominal α on 200 streams per level. Points lie on or below the $y = x$ Ville bound.

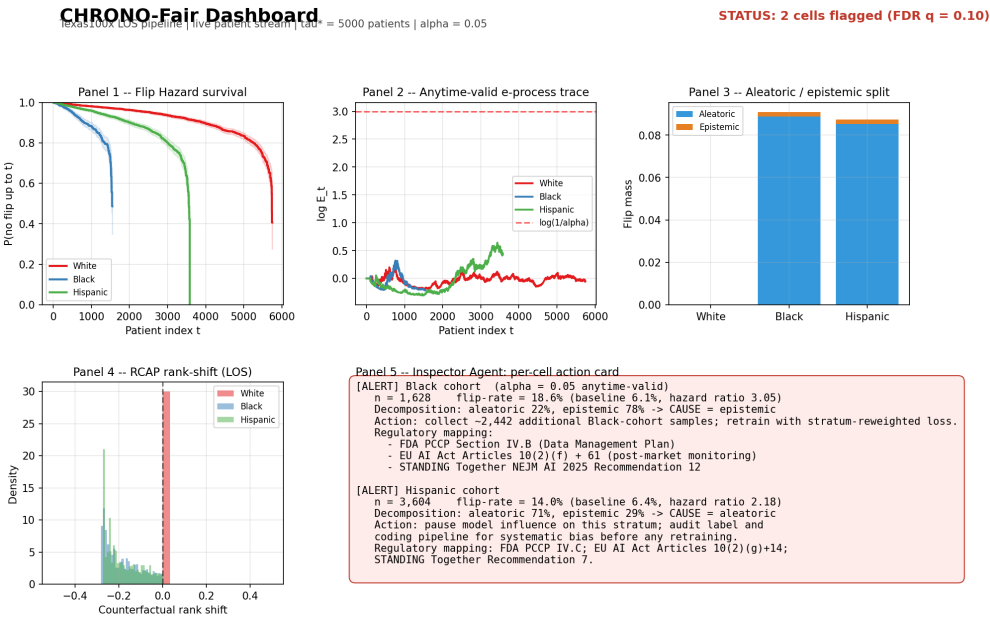


Fig. 10. Section 6.7. Static rendering of the four-panel dashboard at $n = 12000$ with drift onset at $t = 6000$. Inspector Agent card lists cell labels, e-values, hazard ratios, dominant cause, recommended action, and suggested regulatory mapping.

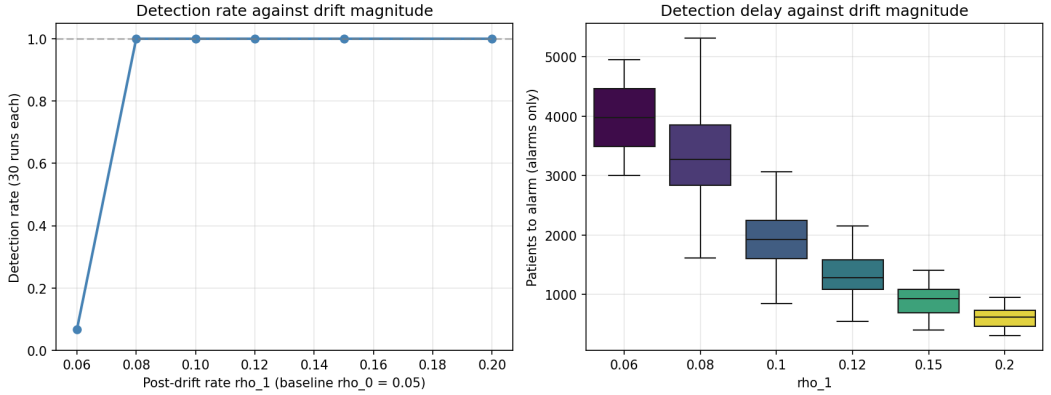


Fig. 11. Section 6.9. Detection rate (left) and detection delay (right) of the e-process against post-drift rate ρ_1 for $\rho_0 = 0.05$ and 30 streams per setting.

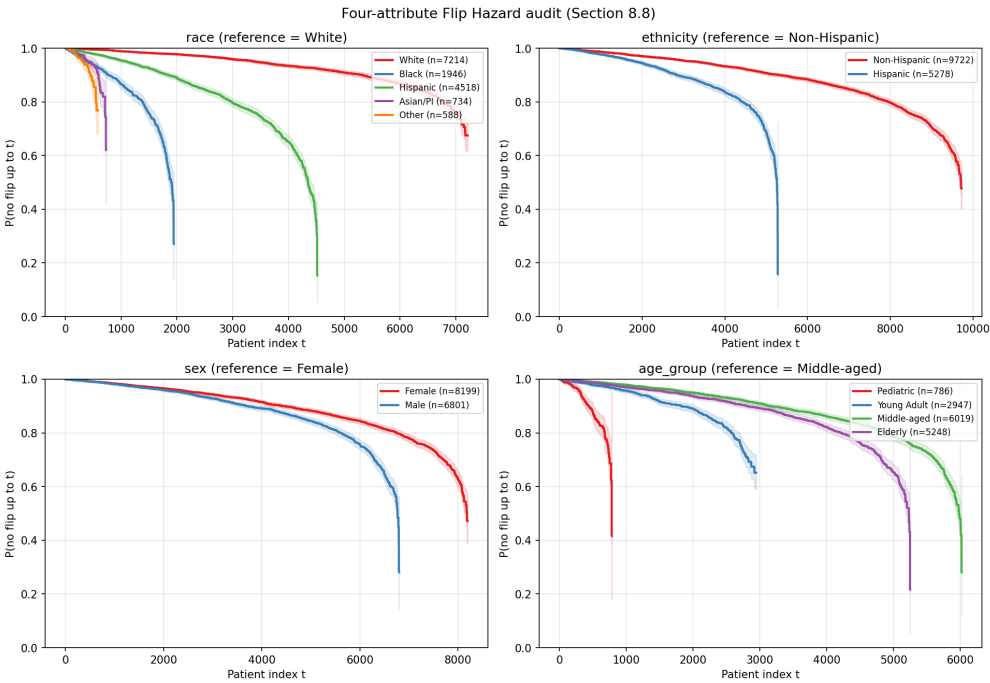


Fig. 12. Section 6.10. Four-attribute Flip Hazard audit. Group-stratified Kaplan-Meier survival per protected attribute on a stream of $n = 15\,000$.

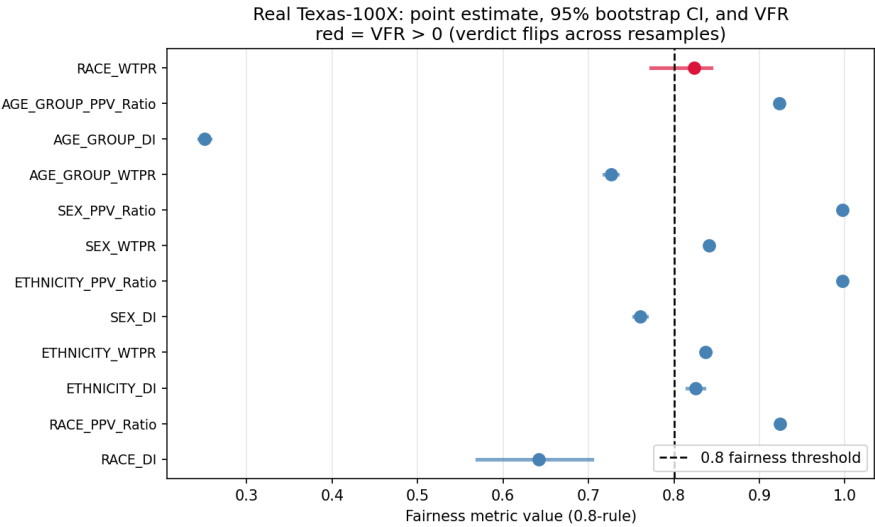


Fig. 13. Section 6.8. Real Texas-100X verdict stability. Point estimate (marker) and 95% bootstrap confidence interval (bar) for 12 metric-attribute combinations, against the 0.8 fairness line. Red marks the one combination with a non-zero Verdict Flip Rate, race worst-group true-positive rate, where the CI straddles 0.8 and 4 of 20 resamples flip the verdict.